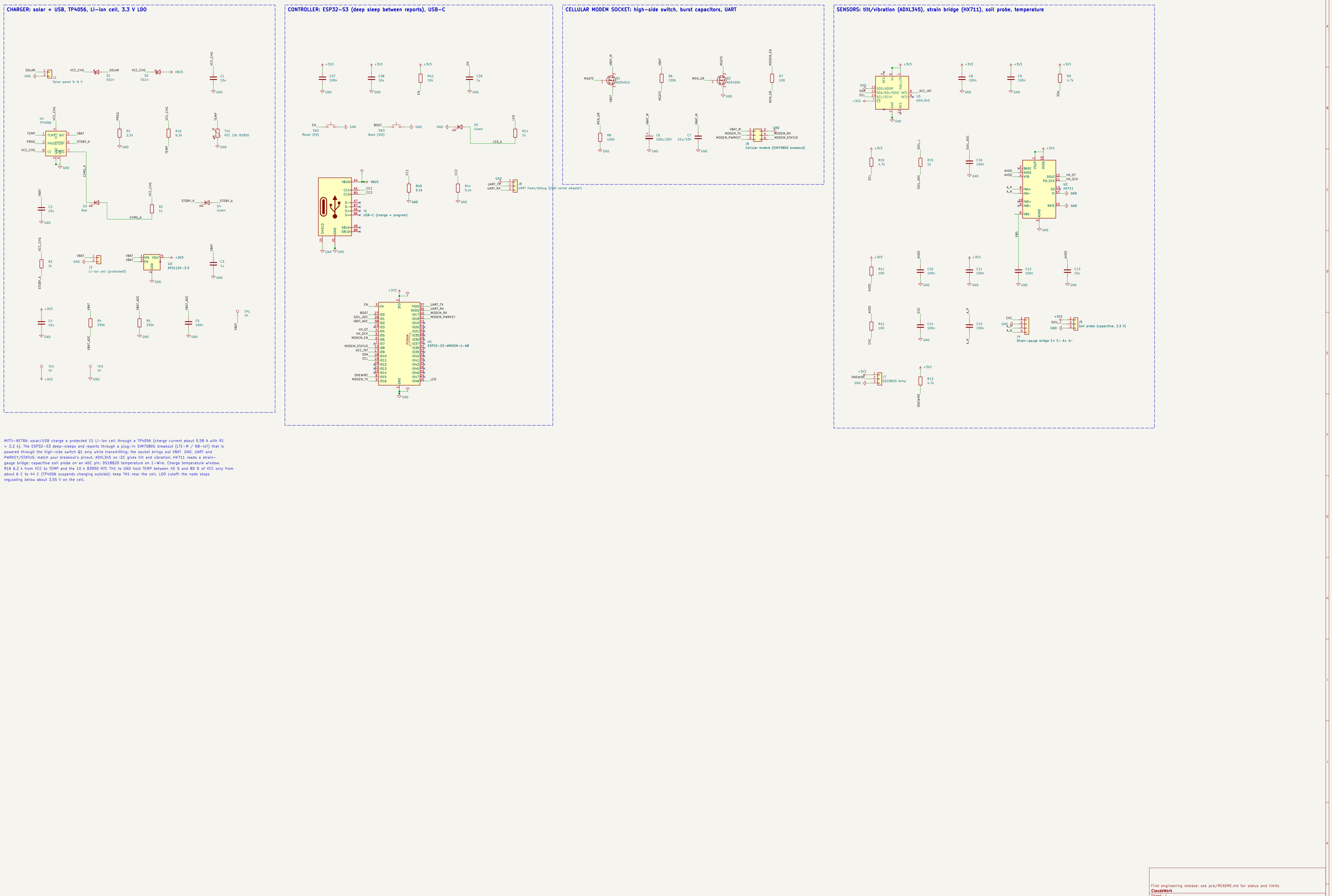


MITTI-NETRA: solar/USB charge a protected 1S Li-ion cell through a TP4056 (charge current about 0.58 A with R1 = 2.2 k). The ESP32-S3 deep-sleeps and reports through a plug-in SIM7080G breakout (LTE-M / NB-IoT) that is powered through the high-side switch Q1 only while transmitting; the socket brings out VBAT, GND, UART and PWRKEY/STATUS; match your breakout's pinout. ADXL345 on I2C gives tilt and vibration; HX711 reads a strain-gauge bridge; capacitive soil probe on an ADC pin; DS18B20 temperature on 1-Wire. Charge temperature window: R16 6.2 k from VCC to TEMP and the 10 k B3950 NTC TH1 to GND hold TEMP between 45 % and 80 % of VCC only from about 6 C to 44 C (TP4056 suspends charging outside); keep TH1 near the cell. LDO cutoff: the node stops regulating below about 3.55 V on the cell.

First engineering release; see pcb/README.md for status and limits

ClaudeWork

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Title: MITTI-NETRA soil and structure telemetry node (ESP32-S3 + LTE-M m

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